

Jianlei Huang

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INFORMATION

Website: <https://jimmy000207.github.io/>

EDUCATION

Fudan University

Shanghai

Bachelor of Engineering in Microelectronic Science and Engineering

Sept. 2018-June 2023

Relevant courses: *Mathematical Analysis (3 semesters); Linear Algebra; Advanced Algebra (2 semesters); Functions of Complex Variable; Probability and Mathematical Statistics (Honorable Level)*

Georgetown University

Washington, DC

Master of Science in Mathematics and Statistics

Aug. 2023-May 2025

Relevant courses: *Real Analysis (Ph.D. Level); Complex Variables (Ph.D. Level); Harmonic Analysis (Ph.D. Level); Partial Differential Equations (Ph.D. Level); Geometric Analysis (Ph.D. Level); Numerical Analysis (Ph.D. Level); Intro Partial Differential Equations; Probability; Mathematical Statistics; Deterministic Math Models*

Georgetown University

Washington, DC

Ph.D. in Applied Mathematics

Aug. 2025-Present

RESEARCH & PROJECT EXPERIENCE

Well-posedness for the Cubic Nonlinear Schrödinger Equation With Nonlinearity Supported on a Hypersurface

Supervised by Professor Benjamin Harrop-Griffiths from Georgetown University

Sept. 2025-Present

- Derived a suitable Strichartz estimate for our equation
- Proved a local well-posedness result in anisotropic Sobolev spaces on $\mathbb{R}^n$ using the contraction mapping
- proved the global well-posedness in the energy space for the defocusing equation in dimension 2 using dispersive estimates and energy conservation.
- Paper [1] in preparation.

Gaussian Priors for Boundary Value Problems of Linear Partial Differential Equations

Collaborated Work with Prof. Bogdan Raiță from Georgetown University

Dec. 2023-Nov. 2024

- Developed a new basis for the solution space derived from the Ehrenpreis-Palamodov Theorem
- Provided formal proofs of correctness.
- Provided empirical results demonstrating significant accuracy and computational resource improvements over state-of-the-art approaches.
- Paper [2] is currently under review at ICRL 2026.

On Non-symmetric Heisenberg Group with in High Dimension

Tutorial research with Prof. Der-Chen Chang from Georgetown University

Sept. 2025-Present

- Derived the Sub-Laplacian and solved the Hamilton-Jacobi equations.
- Derived and counted geodesics.

PUBLICATIONS & PREPRINTS

- [1] [Jianlei Huang](#) and X. Chen. Well-posedness for the cubic nonlinear schrödinger equation with nonlinearity supported on a hypersurface. *In preparation*, 2025.
- [2] [Jianlei Huang](#), M. Härkönen, M. Lange-Hegemann, and B. Raiță. Gaussian process priors for boundary value problems of linear partial differential equations. [arxiv:2411.16663](https://arxiv.org/abs/2411.16663). *Submitted*, 2025.

TEACHING

TA, Georgetown University

- MATH 3320: Functions of Complex Variables. Spring 2026.
- MATH 2410: Ordinary Differential Equations. Fall 2025.
- MATH 8130: Complex Variables. Spring 2025.
- MATH 8100: Real Analysis. Fall 2024.
- MATH 2430: Convex Geometry, Functions & Applications. Spring 2024.

INVITED TALK

APMA Doctoral Seminar	Georgetown University, Washington, D.C.
Topic: GP for Boundary Value Problems of Linear Partial Differential Equations	Oct. 11, 2024

CONFERENCE ATTENDED

AMS 2024 Spring Eastern Sectional Meeting	Apr. 6-7, 2024
Mid-Atlantic Regional Math Alliance conference	Apr. 20, 2024
Prairie Analysis Seminar 2024	Oct. 25-26, 2024
Prairie Analysis Seminar 2025	Nov. 7-8, 2025
Potomac Regional Analysis Seminar	Weekly, Fall 2024

AWARDS & TECHNICAL PROFICIENCIES

● 2021 Interdisciplinary Contest in Modeling: <i>Finalist (top 1%)</i>	<i>Feb. 2021</i>
● 2020 China University Mathematics Contest in Modeling: <i>Second Prize</i>	<i>Sept. 2020</i>
● 2020 Interdisciplinary Contest in Modeling: <i>Meritorious Winner (top 5%)</i>	<i>Feb. 2020</i>
● Fudan University COVID-19 Mathematical Modeling Contest: <i>Second Prize</i>	<i>Jan. 2020</i>
● Technical skills: Latex, Python, R, C++, C, MATLAB	
● Languages: Mandarin (native), English (proficient)	